PRODUCT USER MANUAL

For the two Baltic Sea Products:

Biogeochemical Multi Year Product

BALTICSEA_MULTIYEAR_BGC_003_012

&

Physical Multi Year Product

BALTICSEA_MULTIYEAR_PHY_003_011

Issue: 1.0

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	2022/11	All	New document including both the PHY and BGC reanalysis products: BALTICSEA_ MULTIYEAR_PHY_003_011 & BALTICSEA_ MULTIYEAR_BGC_003_012	Ida Ringgaard, Vasily Korabel, Lena Spruch, Anja Lindenthal, Vibeke Huess	Vibeke Huess
	2023/02	IV	Corrected Product filenames	Vibeke Huess	
	2024/06		Minor update to include operational temporal extensions	Vibeke Huess	

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GLOSSARY AND ABBREVIATIONS

MFC	Monitoring and Forecasting Centre
NetCDF	Network Common Data Form
CF	Climate Forecast (convention for NetCDF)
BAL	Baltic
CF	Climate Forecast (convention for NetCDF)
PC	Production Center
PU	Production Unit
ftp	Protocol to download files
OpenDAP	Open-Source Project for a Network Data Access Protocol. Protocol to download subset of data from a n-dimensional gridded dataset (ie: 4 dimensions: lon-lat,depth,time)
Subsetter	Copernicus Marine service tool to download a NetCDF file of a selected geographical box using values of longitude and latitude, and time range

I INTRODUCTION

I.1 Summary

This document presents a detailed description of the 2 products:

- BALTICSEA_ MULTIYEAR_BGC_003_012
- BALTICSEA_ MULTIYEAR_PHY_003_011

with respect to the available data services and how to access them, as well as how to use the files and services.

These two products provides the multi-year products for the biogeochemical and the physical conditions in the Baltic Sea provided by the Copernicus Marine Service Baltic Monitoring and Forecasting Centre (BAL MFC). The two products are described together in this User Manual as the production of the hindcast was run as one online coupled system.

The product BALTICSEA_ MULTIYEAR_PHY_003_011 is based on simulations with the 3D ocean-ice model Nemo (Nucleus for European Modelling of the Ocean), which is an open community ocean model (https://nemo-ocean.eu). In the BAL MFC model system the Nemo version 4.0 is used in combination with the sea ice and thermodynamic model SI3. Our Nemo configuration and set-up is adapted for the Baltic Sea.

The product BALTICSEA_ MULTIYEAR_BGC_003_012 is based on simulations with the biogeochemical model ERGOM (Ecological Regional Ocean Model, https://ergom.net/), which is developed at IOW (Germany) to model the biogeochemical cycles in the Baltic Sea. The version used in the BAL MFC model system is based on Neumann et al., 2015: *A new radiation model for Baltic Sea ecosystem modelling. Journal of Marine Systems 152: 83-91.* The model describes the basic nutrient and carbon cycles through 15 pelagic state variables. The sediment is not vertically resolved and consists of two organic nutrient components. This product has been produced as a one-way online coupled run with the BAL MFC's Nemo system and set up.

The product BALTICSEA_ MULTIYEAR_BGC_003_012 provides these data:

- Nutrients (phosphate, nitrate and ammonium),
- Oxygen (dissolved oxygen and bottom oxygen),
- Plankton (chlorophyll-a),
- Primary production (net primary production),
- Optical properties (Secchi depth) and
- Carbon (surface partial pressure of CO2, and ph).

All variables are stored as 3D values with up to 56 levels, except the Secchi depth, the surface partial CO2 pressure and the bottom oxygen, which are stored as 2D values.

Data are provided in these 4 different dataset:

- One dataset with daily mean averaged values of the above parameter list. The Secchi depth is inlcuded as instantaneous values from noon UTC ; and the net primary production is stored as accumulated daily values.
- One dataset with monthly mean averaged values of the above parameter list.
- One dataset with annual mean averaged values of the above parameter list.
- One dataset with static files with coordinate and bathymetry information.

The product BALTICSEA_ MULTIYEAR _PHY_003_011 contains model calculations for the physical condition in the Baltic Sea. Data are served in these 4 different dataset:

- One dataset with daily mean averaged values for the parameters: sea surface height variations, sea ice concentrations and thickness at the surface and temperature, salinity and horizontal velocities for 56 vertical levels. Further the temperature and salinity at the bottom layers is provided as well as the depth of the mixed layer.
- One dataset with monthly mean averaged values of the above parameter list
- One dataset with annual mean averaged values of the above parameter list
- One dataset with static files: with coordinate, bathymetry and Mean Dynamic Topography information.

If you have any questions to the products, please contact us.

I.2 History of changes

Date	Description of changes and impacted product
2023/03/30	Both products are introduced as new products in the Copernicus Marine Service product Catalogue in March 2023. They supersede the two old products: BALTICSEA_REANALYSIS_BIO_003_012 BALTICSEA_REANALYSIS_PHY_003_011
2024/06	The two products are intended to be extended in time twice per year; summer and winter with 6 new months for providing data as close as possible to present.

II DESCRIPTION OF THE PRODUCT SPECIFICATION

II.1 General Information

Product Lines	BALTICSEA_MULTIYEAR_BGC_003_012
Geographical coverage	9°E→ 30°E ; 53°N → 66°N
Variables	• Chlorophyll-a (chl) • Nitrate (no3) • Phosphate (po4) • Ammonium (nh4) • Dissolved Oxygen (o2) • Bottom Dissolved Oxygen (o2b) • Ph (ph) • Partial CO2 pressure at the surface (spco2) • Secchi Depth (zsd) • Net primary production (nppv)
Product Type	Multi year
Available time series	From 1/1 1993 and as close as possible to present (approximately op to – 0.5-1 year to present time).
Temporal resolution	Daily mean values. Monthly mean values. Annual mean values.
Target delivery time	N/A
Delivery mechanism	Subsetter and FTP
Horizontal resolution	App. 1 nautical miles: Delta Longitude: 1’40’’ or 0.028 degrees. Delta Latitude: 1’ or 0.017 degrees
Number of vertical levels	Up to 56 levels.
Format	Netcdf & CF1.0

Table 1: BAL-MFC BGC Multi Year product

Product Lines	BALTICSEA_MULTIYEAR_ PHY_003_011
Geographical coverage	9°E→ 30°E ; 53°N → 66°N
Variables	• Sea surface variations (sla) • Ice concentration (siconc) and thickness (sithick) • Temperature (thetao) • Salinity (so) • Horizontal velocities (uo eastward and vo northward) • Mixed Layer Depth (mlotst) • Bottom Temperature (bottom) • Bottom Salinity (sob)
Product Type	Multi year
Available time series	From 1/1 1993 and as close as possible to present (approximately op to – 0.5-1 year to present time).
Temporal resolution	Daily mean values. Monthly mean values. Annual mean values.
Target delivery time	N/A
Delivery mechanism	Subsetter and FTP
Horizontal resolution	App. 1 nautical miles: Delta Longitude: 1’40’’ or 0.028 degrees Delta Latitude: 1’ or 0.017 degrees
Number of vertical levels	Up to 56 levels
Format	Netcdf & CF1.0

Table 2: BAL-MFC PHY Multiyear product

II.2 Details of datasets

BALTICSEA_MULTIYEAR_BGC_003_012
Dataset: **cmems_mod_bal_bgc_my_P1D-m** contains: Daily mean values (24 hours average) of all the below listed parameters (except nppv and zsd) The Secchi Depth (zsd) values are instantaneous values valida at noon UTC. The net primary production (nppv) values are daily accumulated values. **cmems_mod_bal_bgc_my_P1M-m** contains: Monthly mean values of all the below listed parameters **cmems_mod_bal_bgc_my_P1Y-m** contains: Annual mean values o of all the below listed parameters **The NetCDF files contains: variable name and unit; Long_name; CF Standard_name.**
chl [mg /m3] Chlorophyll_a_Concentration mass_concentration_of_chlorophyll_a_in_sea_water
nh4 [mmol /m3] Ammonium_Concentration mole_concentration_of_ammonium _in_sea_water
no3 [mmol /m3] Nitrate_Concentration **mole_concentration_of_nitrate_in_sea_water**
nppv [mg /(m3 day)] net_primary_production_of_biomass_expressed_as_carbon_per_unit_volume net_primary_production_of_biomass_expressed_as_carbon_per_unit_volume_in_sea_water
o2 [mmol/m3] Dissolved_Oxygen_Concentration **mole_concentration_of_dissolved_molecular_oxygen_in_sea_water**
o2b [mmol /m3] Sea floor Dissolved_Oxygen_Concentration **mole_concentration_of_dissolved_molecular_oxygen_in_sea_water**
ph [1] sea_water_ph sea_water_ph_reported_on_total_scale
po4 [mmol /m3] Phosphate_Concentration mole_concentration_of_phosphate_in_sea_water
spCO2 [Pa] surface_partial_pressure_of_carbon_dioxide surface_partial_pressure_of carbon_dioxide_in_sea_water
zsd [m] Secchi_depth

secchi_depth_of_sea_water
Dataset: **cmems_mod_bal_bgc_my_static contains the below 2 files.** **The NetCDF files contains: variable name and unit; Long_name; CF Standard_name.**
File: BAL-MFC_003_012_coordinates.nc: **e1t [m]** Cell dimension along X axis **e2t [m]** Cell dimension along Y axis **e3t [m]** Cell dimension along Z axis cell_thickness
File: BAL-MFC_003_012_mask_bathy.nc: **mask [1]** Land-sea mask: 1 = sea ; 0 = land sea_binary_mask **deptho [m]** Bathymetry sea_floor_depth_below_geoid **deptho_lev [1]** Model level number at sea floor model_level_number_at_sea_floor

Table 3: list of the datasets and variable names and unit for the BALTICSEA_MULTIYEAR_BGC_003_012 product

BALTICSEA_MULTIYEAR_PHY_003_011
Dataset: **cmems_mod_bal_phy_my_P1D-m** contains: Daily mean values (24 hours average) of all the below listed parameters **cmems_mod_bal_phy_my_P1M-m** contains : Monthly mean values of all the below listed parameters **cmems_mod_bal_phy_my_P1Y-m** contains : Annual mean values of all the below listed parameters **The NetCDF files contains: variable name and unit; Long_name; CF Standard_name.**
bottomT [degree_C] Sea floor potential temperature sea_water_potential_temperature_at_sea_floor
mlotst [m] Ocean mixed layer thickness defined by density (as in de Boyer Montegut, 2004) ocean_mixed_layer_thickness_defined_by_sigma_theta
siconc [0, 1] Sea ice cover sea_ice_area_fraction
sithick [m] Sea ice thickness sea_ice_thickness
sla [m] Sea level anomalies sea_surface_height_above_sea_level
so [1e-3] Salinity sea_water_salinity
sob [1e-3] Sea water salinity at sea floor sea_water_salinity
thetao [degree_C] Potential temperature sea_water_potential_temperature
u0 [m/s] Eastward current eastward_sea_water_velocity
v0 [m/s] Northward current northward_sea_water_velocity
Dataset: **cmems_mod_bal_phy_my_static contains the below 3 files.** **The NetCDF files contains: variable name and unit; Long_name; CF Standard_name.**

File: BAL-MFC_003_011_coordinates.nc: **e1t [m]** Cell dimension along X axis **e2t [m]** Cell dimension along Y axis **e3t [m]** Cell dimension along Z axis **cell_thickness**
File: BAL-MFC_003_011_mask_bathy.nc: **mask [1]** Land-sea mask: 1 = sea ; 0 = land sea_binary_mask **deptho [m]** Bathymetry sea_floor_depth_below_geoid **deptho_lev [1]** Model level number at sea floor model_level_number_at_sea_floor
File: BAL-MFC_003_011_mdt.nc: **mdt [m]** Mean dynamic topography sea_surface_height_above_geoid

Table4: list of the datasets and variable names and unit for the BALTICSEA_MULTIYEAR_PHY_003_011 product

II.3 Production System Description

II.3.1 Model system

The BAL MFC model system consist of four building blocks:

(1) The **3D ocean-ice model Nemo** (Nucleus for European Modelling of the Ocean), which is an open community ocean model (https://nemo-ocean.eu). In the BAL MFC model system the Nemo version 4.0 is used in combination with the sea ice and thermodynamic model SI3. Our Nemo configuration and set-up is adapted for the Baltic Sea.

(2) The **biogeochemical model ERGOM** (Ecological Regional Ocean Model, https://ergom.net/), which is developed at IOW (Germany) to model biogeochemical cycles in the Baltic Sea. The version used in the BAL MFC model system is based on Neumann et al., 2015*: A new radiation model for Baltic Sea ecosystem modelling. Journal of Marine Systems 152: 83-91.* The model describes the basic nutrient and carbon cycles through 15 pelagic state variables. The sediment is not vertically resolved and consists of two organic nutrient components.

(3) The **wave model WAM**, an open source model developed at Hereon (Germany). The WAM version used in the BAL MFC model system is based on WAM version 4.6.2., and has been adapted to the Baltic Sea. The model uses 36 spectral directions and 35 different frequencies between 0.04177 and 1.06719 Hz.

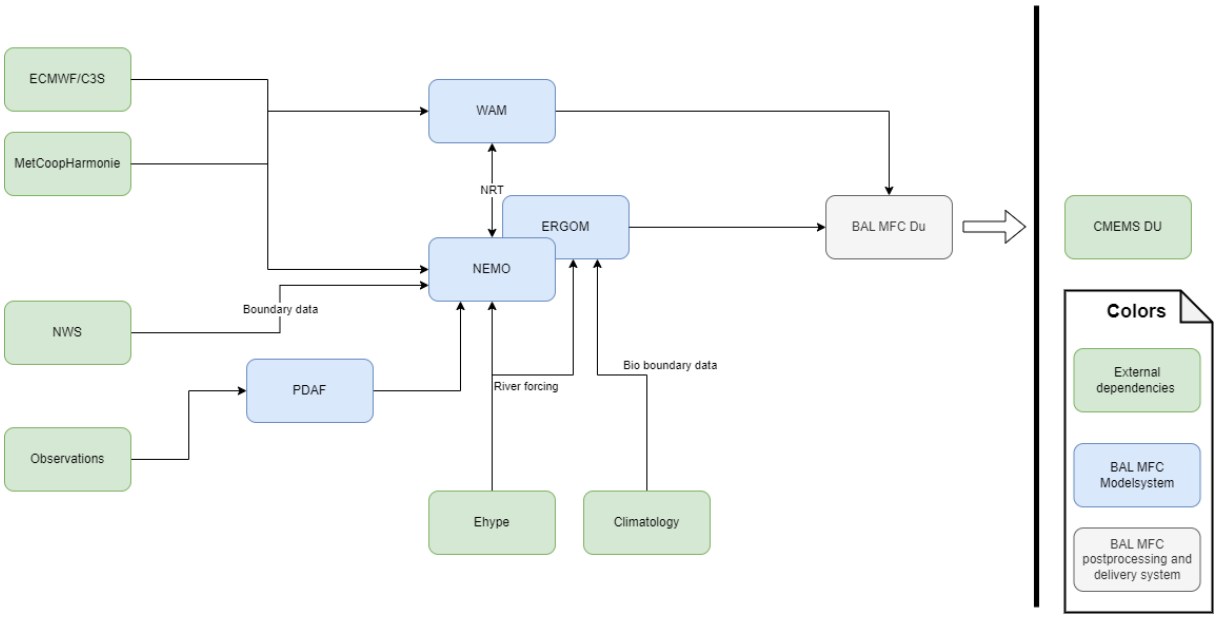
(4) The **data assimilation system PDAF** (Parallel Data Assimilation Framework https://pdaf.awi.de/), used with the LESTKF filter.

The Nemo and ERGOM models are run one-way online-coupled and use the PDAF system to assimilate observations into the analysis field in Nemo. The WAM model runs independently from the Nemo-ERGOM-PDAF system. Figure 1 shows an overview of the complete BAL MFC model system set-up.

Both model systems run with a horizontal resolution of 1 nautical mile and use the same bathymetry grid based on GEBCO v2013 data. Both model systems are forced with the same best available atmospheric forcing dataset.

The native model domain for the Nemo-ERGOM system covers the North Sea and Baltic Sea with open boundaries at Scotland-Norway and the English Channel. The wave system's native model domain covers the Baltic Sea and towards the eastern part of the North Sea (from 8 degree E).

Our production with the Nemo v.4.0 system is using the TEOS-10 equation of state for the model calculations. The temperature and salinity input data are converted to absolute salinity and conservative temperature, and the final products are calculated back to practical salinity and potential temperature.



II.3.2 Multi year (MY) production system

The BAL MFC multi year products (BALTICSEA_MULTIYEAR_PHY_003_011 and BALTICSEA_MULTIYEAR_BGC_003_012) are produced with the online one-way coupled Nemo-ERGOM model. The BAL MFC multi year product for the wave conditions (BALTICSEA_REANALYSIS_WAV_003_015) is produced with the WAM system. These two multi year production systems are for the time being run without any interaction (i.e. with no exchange of parameters). The ERA5 dataset available from Copernicus Climate Service is used as atmospheric forcing data for both production systems. The products cover the period from 1/1 1993 up till as close to present as possible (approximately present time minus 1-1.5 Y).

Details for the Nemo-ERGOM production:

The following parameters are used as atmospheric forcing: 10 m wind speed and direction, surface temperature, air pressure at sea surface, solar radiation total precipitation, air humidity. For the atmospheric deposition of nitrate and ammonium, ERGOM uses the yearly data set from EMEP (https://emep.int/mscw/mscw_moddata.html). The atmospheric concentration of CO_2 is estimated from the NOAA global monthly mean CO_2 data (https://gml.noaa.gov/ccgg/).

Information about river discharges and the rivers water temperature are included in the Nemo-ERGOM system from the historical data set of the EHYPE-model system (version: ehype3_fgd produced at SMHI, Sweden in 2020). This data set also contains nutrient loads for nitrogen (inorganic and total) and phosphorus (soluble reactive and total) used by ERGOM. River loads of silicate, dissolved inorganic carbon and alkalinity are based on Pätsch and

Lenhart (2004) and the Baltic database. Data from 849 rivers are included (for the native North Sea – Baltic Sea full domain).

At the lateral boundaries, daily mean values from the CMEMS North West Shelf multi-year product are used (NWSHELF_MULTIYEAR_PHY_004_009, 10.48670/moi-00059). From this model, the following parameters are used: Sea level, temperature, salinity, horizontal velocities. Additionally, hourly tidal information is added at the open boundaries based on 9 tidal constituents from a tidal model.

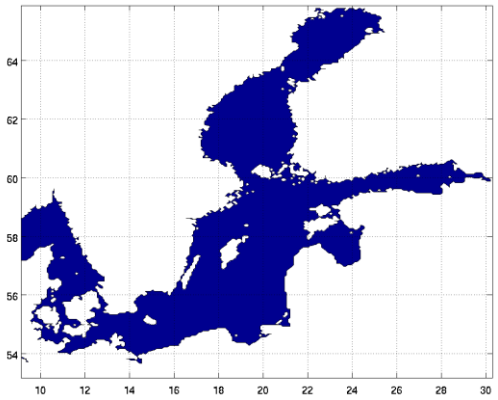
For the biogeochemical model ERGOM interpolated data from ICES for DIN, DIP, silicate and oxygen are used at the lateral boundaries (https://www.ices.dk). Zoo- and phytoplankton data are based on ICES observations of chlorophyll-a (chlorophyll to plankton biomass). The parameters of the carbonate system Total Alkalinity (TA) and Dissolved Inorganic Carbon (DIC) are based on Wolf-Gladrow et al. (2007, https://doi.org/10.1016/j.marchem.2007.01.006) and results of the biogeochemical model ECOHAM (Lorkowski et al., 2012, https://doi.org/10.1016/j.ecss.2011.11.037).

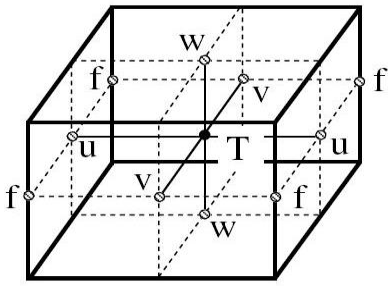
Satellite observations of SST and in-situ profile observations are assimilated into the Nemo analysis field using the PDAF tool at every midnight. The assimilation is done in a three day cycle, alternating between the assimilation of SST, temperature profiles and salinity profiles. All variables are assimilated separately in a uni-variate sense. Increments are added to the model forecast state using Incremental Analysis Update (IAU) module with the time window of 24 hours. The satellite observation are based on the Copernicus Marine Service L3S SST multi year observations. The in-situ profile observations are from the ICES database. A climatologic ensemble with 75 members is used for the assimilation. The ensemble is based on a 5 year (2011-2015) free run of the Nemo system and created with a 15-day moving window.

Summary of the operational model setups: Domain

Resolution and grid

Geographic coverage: North Sea – Baltic Sea **:** 53°N - 66°N, 9°E - 30°E





The horizontal grid step is regular in latitude and longitude and is of approximately 1 nautical miles (lat: 0.017 degrees; lon: 0.028 degrees). The grid is a staggered Arakawa C-grid (see figure 1). The dataset of the scalar variables (temperature, salinity, sea level, ice, and biogeochemical variables) are centred in the T point of the grid; at the grid cell node. The velocity components are located half-way between adjacent grid cell nodes on the face of the grid cell in the respective direction. The velocities are de-staggered/interpolated to the T point in the delivered products.

Bathymetry	GEBCO at 1 nautical mile horizontal resolution
Vertical grid	The product is delivered at the models native grid, with up to 56 vertical levels: levels [m]: 0.50, 1.52, 2.55, 3.60, 4.68, 5.80, 6.96, 8.17, 9.45, 10.81, 12.27, 13.86, 15.60, 17.53, 19.70, 22.15, 24.94, 28.15, 31.86, 36.15, 41.13, 46.91, 53.59, 61.28, 70.08, 80.07, 91.31, 103.82, 117.62, 132.67, 148.90, 166.25, 184.60, 203.86, 223.92, 244.66, 265.99, 287.81, 310.05, 332.64, 355.51, 378.61, 401.91, 425.36, 448.93, 472.61, 496.36, 520.19, 544.06, 567.98, 591.94, 615.92, 639.92, 663.94, 687.98, 712.02
Model Version **Production units** **Period of validation**	▪ NEMO 4.0 – ERGOM_v202206 ▪ DMI, Denmark & BSH, Germany ▪ The initial validation is performed on this initial product period : 01.11.1993 - 31.12.2021
Assimilation scheme	PDAF LESTKF univariate for SST & T/S profiles

Assimilated observations	SST_BAL_PHY_L3S_MY_010_040 S & T profiles from the ICES and SHARK-databases are used.
Initial conditions (spin-up process)	From the previous BAL MFC Multi Year products (based on the NEMOv3.6 – SCOBI system).
NEMO-ERGOM coupling	The biogeochemical model is online coupled to the physical through an operator splitting scheme where first advection and diffusion of the tracers are calculated in Nemo and thereafter ERGOM calculates the biogeochemical sinks and sources. The biogeochemical processes are updated at every advection timestep in Nemo.

II.4 Grid

The product files are on a regular lat-lon grid and all values are delivered at the Tracer-point (i.e. the 3D velocities are interpolated to the Tracer-points).

II.5 Processing information

II.5.1 Update Time

The two reanalysis products were run in a homogeneous sense (i.e. with homogeneous forcing/input fields) for the initial production performed in 2022 for the data period: 1/1 1993 – 31/12 2021. The two products will be extended in time afterwards using the best available data to be used as forcing/lateral boundary/input fields. The plan is to update the products twice per year; to extend the reanalysis products as close to present time as possible.

III DOWNLOAD A PRODUCT

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

IV FILES NOMENCLATURE

The file nomenclature depends on the download source.

You will find the different nomenclature rules in this article : How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center

However, as the old download protocols have been replaced recently by new ones (subset, files and maps), some files may be still under old nomenclature. You will find them below.

IV.1 Nomenclature of files when downloaded through the Web Portal Subsetter Service

Files nomenclature when downloaded through the Copernicus Marine Web Portal Subsetter is based on product dataset name and a numerical reference related to the request date on the MIS.

The scheme is: **datasetname_ nnnnnnnnnnnnn.nc**

where :

- ✓ **datasetname**: as described previously
- ✓ **nnnnnnnnnnnnn**: 13 digit integer corresponding to the current time (download time) in milliseconds since January 1, 1970 midnight UTC.
- ✓ **.nc**: standard NetCDF filename extension.

IV.2 Nomenclature of files when downloaded through the FTP Service

BALTICSEA_MULTIYEAR_BGC_003_012:
file nomenclature when downloaded through the Copernicus Marine FTP service is based on the original file names behind the products. The files are in NetCDF and are named:

- For the daily mean values:
  BAL-MYP-ERGOM_BGC-DailyMeans-“ccyymmdd”.nc
  Each file contains 1 timestamp for the calendar day.

- For the monthly mean values:
  BAL-MYP-ERGOM_BGC-MonthlyMeans-“ccyymm”.nc
  Each file contains 1 timestamp for the calendar month.

- For the annual mean values:
  BAL-MYP-ERGOM_BGC-AnualMeans-‘ccyy”.nc

Each file contains 1 timestamp for the calendar year.

BALTICSEA_MULTIYEAR_PHY_003_011:

file nomenclature when downloaded through the Copernicus Marine FTP service is based on the original file names behind the products. The files are in NetCDF and are named:

- For the daily mean values:
  BAL-MYPNEMO_PHY-DailyMeans-“ccyymmdd”.nc
  Each file contains 1 timestamp for the calendar day.

- For the monthly mean values:
  BAL-MYPNEMO_PHY-MonthlyMeans-“ccyymm”.nc
  Each file contains 1 timestamp for the calendar month.

- For the annual mean values:
  BALMYP-NEMO_PHY-AnnualMeans-“ccyy”.nc
  Each file contains 1 timestamp for the calendar year.

IV.3 Other information: mean centre of Products, land mask value, missing value

Fill and land value: -999

All files are in NetCDF4 format.

IV.4 File size

BALTICSEA_MULTIYEAR_BGC_003_012:
Daily/Monthly/Annual files: 70 Mb/file

BALTICSEA_MULTIYEAR_PHY_003_011:
Daily/Monthly/Annual files: 45 Mb/file

V FILE FORMAT

The products are stored using the NetCDF format.

To know more about the formats, please follow this link:

What are the formats of Copernicus Marine products? | Copernicus Marine Help Center